Lab 10A: Repeated Measures.

Question 1. Nested Designs:

In this assignment, we will work on partitioning the variance in a nested design from the textbook. Using the data in the table below, we will need to partition the variance in our data in order to appropriately analyze the independent components of our data. Fill in the missing cells in the table below.

Note that each of these answers comes the problem being solved in the lecture. The formulas to get these numbers are supplied in the power-points for that lecture.

If you have any questions or confusion about the formulas used to get these answers, please send me an email so that we can set up a time to meet.

These formulas are challenging, so I certainly expect people to have questions. :)

Instructions (j)

	Competition			**No Competition**		
	Group 1	**Group 2**	**Group 3**	**Group 4**	**Group 5**	**Group 6**
Y_{hik}	8	6	7	6	7	5
Y_{hik}	9	7	8	6	6	5
Y_{hik}	7	6	8	5	7	4
Group Mean	8	6.33	7.67	5.67	6.67	4.67
Condition Mean	7.33			5.67		
Grand Mean	6.5					

Next, using the formulas provided in the .pdfs, complete the ANOVA table below.

Source	SSE	df	MSE	F_obs	PRE
Condition	12.5	1	12.50000	4.68	0.54
Error Between Groups	10.67	4	2.667500		
Total Between Groups	23.17	5	4.634000		
Total Within Group	5.33	12	0.444167		
Total	28.5	17			

Finally, we want to compute the ***intra-class correlation coefficient***. This correlation tells us the degree to which scores from same group resemble each other (i.e., are observations more similar within groups than between groups) to calculate this, we will need the MSE for the error between groups, the MSE total within groups, and the number of observations per group (labelled 's'):

$$ICC = \frac{MSE_b - MSE_w}{MSE_b + MSE_w(s-1)} = \frac{2.67-0.44}{2.67+0.44(3-1)} = 0.63$$

Question 2. Crossed Designs:

Next, we will work on partitioning the variance in a crossed-design from the textbook. Using the data in the table below, we will need to partition the variance in our data in order to appropriately analyze the independent components of our data. Fill in the missing cells in the table below.

Note that each of these answers comes the problem being solved in the lecture. The formulas to get these numbers are supplied in the power-points for that lecture.

If you have any questions or confusion about the formulas used to get these answers, please send me an email so that we can set up a time to meet.

These formulas are challenging, so I certainly expect people to have questions. :)

Conditions

Participant	Experimenter Absent	Experimenter Present	*Participant Average*
Y_{hi}	7	8	7.5
Y_{hi}	5	5	5
Y_{hi}	6	6	6
Y_{hi}	7	9	8
Y_{hi}	8	8	8
Y_{hi}	7	7	7
Y_{hi}	5	6	5.5
Y_{hi}	6	8	7
Condition Means	6.375	7.125	6.75
Grand Mean	6.75		

Next, using the formulas provided in the .pdfs, complete the ANOVA table below.

Source	SSE	df	MSE	F_obs	PRE
Total Between Subjects	18	7	2.571429		
Condition	2.25	1	2.250000	5.73	0.45
Error Within	2.75	7	0.392857		
Total Within Subjects	5.00	8	0.625000		
Total	23.00	15			

Finally, we want to compute the ***intra-class correlation coefficient***. This correlation tells us the degree to which scores from same group resemble each other (i.e., are observations more similar within groups than between groups) to calculate this, we will need the MSE for the error between groups, the MSE total within groups, and the number of observations per group (labelled 's'):

$$ICC = \frac{MSE_b - MSE_w}{MSE_b + MSE_w(s-1)} = \frac{2.57-0.39}{2.57+0.39(1)} = 0.74$$

Question 3. Identify Different Types of Variables:

It is common for us to have "participant-level" data. That is, either an animal or human participant represents the level of analysis. As such, we often think of variables as between-subjects or within-subjects rather than crossed or nested per se. For each of the experimental designs below, Identify the within-subject variables and the between subject variables.

- An experiment in which 60 people were randomly assigned to either an experimental cooling treatment for muscle soreness, or a placebo control condition. All participants were assessed at a pre-test, an immediate post-test, and a 1-week follow-up.
    - Between: Treatment Type (Experimental or Control)
    - Within: Time (Pre Test, Post Test, Follow Up)
- A study in which military personnel are asked to make tactical decisions about ten different stimuli under different conditions: difficult without time pressure; difficult with time pressure; easy without time pressure; easy with time pressure. The same stimuli are measured in each condition.
    - Between: None, all variables are manipulated within-subjects.
    - Within: Difficulty (High versus Low), Time Pressure (Present versus Absent)… note that stimulus could also be included as a within-subjects variable. However stimuli are usually treated as a random sample and not included in the model itself.
- A longitudinal study in which boys and girls were observed from the ages of 8-14 and their body composition is measured every year. These children are further broken into groups based on their physical activity levels (averaged over time) to create: low physically active, moderately physically active, and high physically active groups. Within each level of physical activity, there are both boys and girls. All participants were measured at all time points.
    - Between: Physical Activity (Low, Moderate, High), Gender (Boys, Girls)
    - Within: Time (day, month, or year in which the measurement occurred)
- Rats are given tests of their memory (using the Morris Water Maze) over six-months of observation (the rats are tested at the end of every month). Half of the rats spend three months in an "enriched environment" (a pen with lots of toys and peers) and then three months in a "sterile" environment (an identically sized pen with a running wheel and straw). The other half of the rats complete these conditions in the opposite order.
    - Between: Order (enriched then sterile or vice versa). Note that "Order" is often omitted as an explicit factor in experiments. If we were interested in order effects, however, then Order would be a between-rats variable.
    - Within: Time (months); Condition (Enriched Environment versus Sterile Environment).
- Your partner has recently started thinking that they are allergic to dairy. To test this hypothesis, you first learn that it can take dairy up to ten days to leave your system, so you devise a schedule whereby they eat dairy (D) or not (ND) with a two week buffer period (B) between observations: D-B-ND-D-B-ND-D-B-ND. Keeping a symptom log allows you to track the number of adverse events in each period.
    - Between: None, this is an "N of 1" trial so there are no between subjects variables.
    - Within: Condition (D, ND, or B)